

The Vanishing Struggle: Whether Generative AI Substitution Forecloses the Occasions for Reasoning in Undergraduate Writing

A cross-disciplinary, process-based study at the University of Liberal Arts Bangladesh

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Draft proposal · 2026

Abstract

The debate over generative AI in student writing has fixed on a question of *output*: do students who use AI write worse on their own? This study asks a different and, I argue, prior question — one about *process*. Writing is valued in education not only for the text it produces but for the thinking it forces: the moments where a writer confronts an objection, abandons a weak claim, searches for evidence and rejects some, and revises under uncertainty. These moments of *productive struggle* are the occasions on which reasoning is exercised. This study asks whether those occasions still occur when a student writes with generative AI, or whether substitutive AI use removes them.

The design is cross-sectional, mixed-methods, and within-person. Thirty to forty undergraduates, stratified across four ULAB departments (Business, Computer Science / Engineering, English and Humanities, and Media Studies) and spanning a range of AI-use habits, each complete two tracked writing sessions: one without AI and one with AI permitted, both written in Google Docs with full version history preserved. After each session, a short structured reflection log captures where the student got stuck, what they changed, and what they chose not to do. A purposive subset of 12–15 students then sit for semi-structured interviews conducted by a researcher with no teaching link to them. The unit of analysis is not the finished essay but the *struggle-event* — a codeable decision-moment during writing — and the central comparison is within each student, between their no-AI and AI-permitted sessions.

The study makes no claim to measure whether reasoning *develops* over time; a single semester cannot show that. What it can show is whether the conditions under which reasoning would develop are present or absent in the act of AI-assisted writing. The concept of *epistemic foreclosure* — the closing-off of a capacity before it forms — is offered as an interpretive frame for the discussion, and a longitudinal successor study is specified as the design that could test it directly. The contribution is threefold: the first process-based, within-person evidence on AI and writing from a Bangladeshi sample; a transferable method for making “productive struggle” observable; and a clearly scoped foundation for the developmental study that must follow.

1. Introduction and problem statement

While teaching Academic English at ULAB’s Centre for Language Studies in early 2026, I noticed a pattern I could not explain by looking at what students handed in. Their essays were competent — clean, organised, argued. But when I asked them, in class, to defend the same argument aloud, many could not say why their evidence supported their claim, could not answer an obvious objection, could not push their own idea one step further than the sentence already on the page. The writing performed a competence the thinking did not seem to back.

The standard way to study this is to measure the *product*: collect students’ independent writing and their AI-assisted writing and compare the two. That comparison is worth making, but it answers a question that is nearly answered before you start. Of course a student’s writing differs with and without a powerful tool in the loop; a calculator produces a “gap” too. The existence of a difference in output tells us little about learning. What matters is what happens *during* the writing — whether the cognitive work that writing is meant to demand still takes place, or whether the tool now does it.

This study reframes the question from product to process. Its object is the moment of *productive struggle* — the effortful, uncertain, generative work of figuring out what you think and defending it — because that is the moment on which reasoning is exercised and, over time, built. My claim is narrow and, I hope, honest: I cannot show in one semester that AI use erodes or forecloses reasoning as a developed capacity. But I can examine, directly and now, whether the *occasions* for that reasoning still occur when a student writes with AI. If those occasions systematically vanish under substitutive use, that is a finding worth having on its own, and it is the necessary groundwork for the longitudinal study that could establish the developmental consequence.

The Bangladeshi context sharpens the question. Peer-reviewed empirical work on generative AI in Bangladeshi higher education is almost absent, and most global evidence is Western, survey-based, and focused on output. Where students arrive at university with underdeveloped writing — a pattern documented at least locally by Rahman, Hossain, and Uddin (2024) — the concern is not that AI erodes an established skill but that it may remove the struggle through which the skill would first form. I take that as motivation for studying the struggle directly, not as a premise this study can itself prove.

2. Conceptual framework

My argument is a chain of four links, and it matters which kind of claim each one is. Two are premises I *import* from established theory; one is the mechanism this study *observes*; and the last is a conclusion I *reason to*, not one the data delivers. The study lives on Link 2 — the observable disappearance of struggle. Foreclosure (Link 4) is the interpretive reading I offer in the discussion, resting on the premises, and testable only by the longitudinal successor described in Section 8.

The Foreclosure Argument: Four Links

Premises are imported (1, 3); the mechanism is observed (2); foreclosure is the reasoned reading (4)

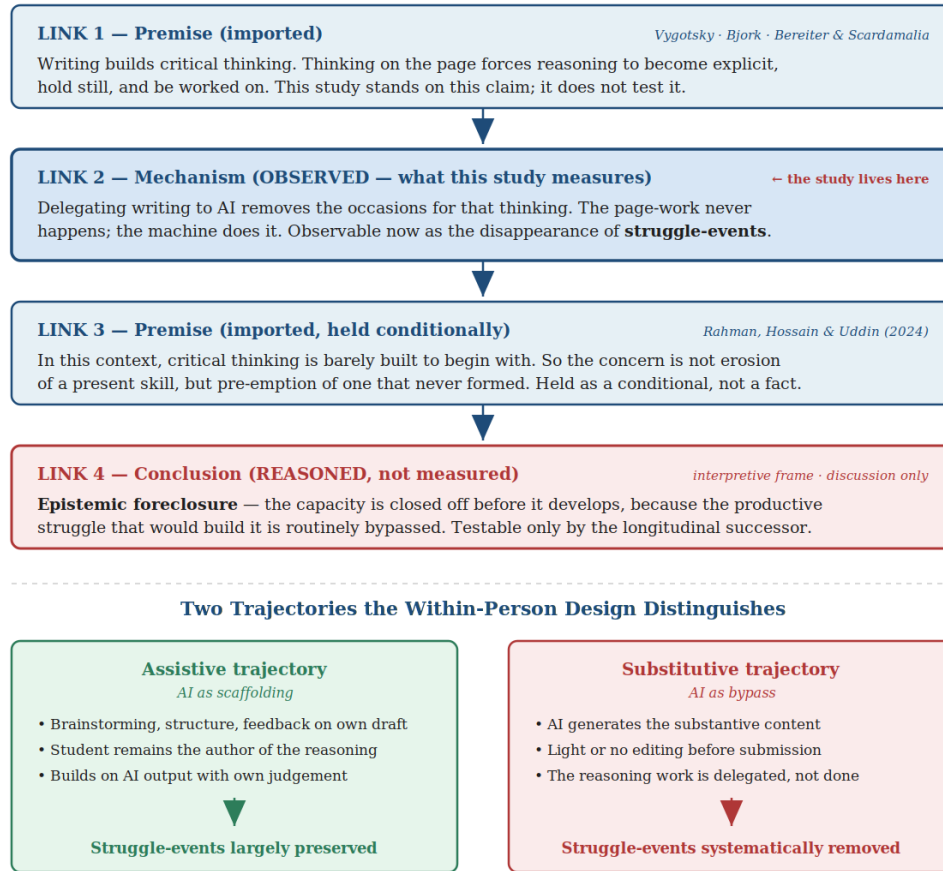


Figure 1. The foreclosure argument as a four-link chain, and the two writing trajectories the within-person design is built to distinguish.

3. Literature and grounding

Writing as thinking

The premise that writing produces knowledge rather than merely recording it runs from Bereiter and Scardamalia's (1987) distinction between knowledge-telling and knowledge-transforming writing, through Flower and Hayes's (1981) account of planning, translating, and reviewing as sites of active reasoning. Transformative writing — engaging with ideas rather than transcribing them — is the mode most associated with higher-order reasoning and the mode most vulnerable to displacement by a system that delivers finished text.

Productive struggle as the mechanism

Vygotsky's (1978) zone of proximal development and Bjork's (1994) *desirable difficulty* converge on a single claim: the features of a task that make it effortful in the moment are the same features that produce durable learning. Remove the difficulty and you remove the mechanism. This study takes

productive struggle not as a metaphor but as its measured construct: the observable moments of effortful, uncertain, generative work during writing.

Cognitive offloading and its limit

Risko and Gilbert (2016) review the mechanisms that trigger cognitive offloading and its immediate performance effects; they do not establish long-term reductions in capacity, and I do not claim they do. Offloading motivates the concern; it does not evidence foreclosure. This is precisely why the present study measures the occasion for reasoning rather than asserting a developmental loss it cannot observe.

The foreclosure frame, honestly bounded

Where writing competence is not yet built, the common “atrophy” framing does not fit — one cannot erode what does not exist. I use *epistemic foreclosure* to name the alternative: the closing-off of a developing capacity when the productive struggle that would build it is routinely bypassed. I treat foreclosure as an interpretive lens for the discussion, not as a hypothesis this design tests. Foreclosure and atrophy predict the same present-tense pattern — fewer struggle-occasions under substitution — and only a longitudinal design can distinguish them. This study measures whether that pattern is present at all.

The gap

No published, peer-reviewed study to my knowledge provides a within-person, process-based comparison of struggle during AI-assisted and independent writing, in any context, let alone a Bangladeshi one. The dominant method is the cross-sectional self-report survey, which cannot see the writing process. This study is an attempt to see it.

4. Research objectives

1. To determine whether, and how much, the occasions for productive struggle during writing differ within the same student between a no-AI and an AI-permitted condition.
2. To characterise the patterns of cognitive delegation and retention — what writing work students hand to AI and what they keep — across a cross-disciplinary sample.
3. To develop and test a transferable method for making productive struggle observable through the combination of document version history, structured reflection, and interview.
4. To specify, on the basis of the findings, the longitudinal design needed to test whether the observed removal of struggle-occasions forecloses reasoning development.

5. Research questions

Primary

- **RQ1.** Within the same student, how do the frequency and type of struggle-events differ between a no-AI writing session and an AI-permitted session?
- **RQ2.** What patterns of cognitive delegation versus retention (assistive versus substitutive use) appear in the AI-permitted sessions, and how do they relate to the change in struggle-events?

Secondary / qualitative

- **RQ3.** How do students describe the experience of writing with and without AI — where they report thinking hardest, where they report stopping, and what they notice themselves delegating?
- **RQ4.** Do struggle-event patterns differ across academic discipline, and if so, how?

6. Method

6.1 Design

Cross-sectional, sequential mixed-methods, with a within-person quasi-experimental core. Each participant writes in two conditions; each participant is their own control. No external or historical baseline corpus is used — the within-person design removes the need for one.

6.2 Sample

Thirty to forty undergraduates, recruited by stratified purposive sampling across four departments — Business Administration (BBA), Computer Science and Engineering / Electrical and Electronic Engineering (STEM), English and Humanities, and Media Studies — and across a self-reported spectrum of AI-use intensity. The cross-disciplinary spread is deliberate: it prevents “reasoning” from collapsing into “writing-major skill.” Recruitment is run by a research assistant outside the lead researcher’s teaching reach, and no participant is drawn from a class the lead researcher teaches, to remove teacher-as-recruiter pressure.

6.3 The two writing sessions (within-person core)

In a single supervised sitting or two closely spaced ones:

- **Session A — no-AI.** A ~40-minute essay (~500 words) on a calibrated prompt, written in a Google Doc with version history on; no AI tools of any kind. Administered in a room, phones away, to make the no-AI condition verifiable rather than merely declared.
- **Session B — AI-permitted.** A comparable essay on a matched prompt, written in a tracked Google Doc, with AI use permitted and a required prompt-and-edit log: what was asked, what was kept, what was changed.

Prompt order and assignment are counterbalanced across participants to separate the AI effect from prompt difficulty and fatigue.

6.4 The reflection log

Immediately after each session, the student completes a short fixed-prompt log (2–3 minutes): where they got stuck, what they changed and why, what they decided *not* to do, and — for Session B — where the tool made a choice for them. The log timestamps the student’s own account of struggle to the session it describes.

6.5 The struggle-event as unit of analysis

A *struggle-event* is a codeable decision-moment during writing: a reformulated claim, an abandoned line of argument, evidence sought and rejected, a counterargument engaged, a substantive revision. Evidence for struggle-events comes from three layers that check one another:

- **Version history** — the objective trace of pauses, deletions, reordering, and rewriting (the shape of the struggle).
- **The reflection log** — the student's account of why those changes happened (the meaning of the struggle).
- **The interview** — the reconstructed experience (the depth).

The version history shows *that* struggle occurred; the log and interview show *what it was*. Triangulation across the three is what makes a small sample credible.

6.6 Interviews

A purposive subset of 12–15 students, chosen to span high and low struggle-visibility in their traces, sit for semi-structured interviews (~45 min) conducted by a third-party researcher with no teaching or evaluative link to them. The guide covers the writing process, what was delegated versus retained, and where the student located their own hardest thinking. Interviews are analysed by reflexive thematic analysis (Braun & Clarke, 2006).

6.7 Coding and reliability

The lead researcher codes all sessions and transcripts against a struggle-event codebook piloted before the main analysis. A second trained coder (a research-literate peer, not a participant and with no stake in the result) independently double-codes a 20% subset; inter-coder reliability is reported as Cohen's κ , with a target of $\kappa \geq .70$. Where reliability falls short after a second training round, the affected code is reported descriptively only.

6.8 Roles (to be confirmed)

Proposed supervisor: Iram Mehrin (CLS); standby: Tazin Ahmed (CLS). Third-party interviewer and second coder: to be confirmed from a research-trained peer pool, contingent on their having no teaching link to recruited participants.

6.9 Ethics

Written informed consent before any data collection; version-history and log data linked to participants by code number only; no session shared with faculty, academic affairs, or grade records under any circumstance; interviews pseudonymised and seen by the lead researcher only after identifiers are removed. Ethics approval sought from the relevant ULAB body before collection begins. Because participation is voluntary and the no-AI condition is supervised rather than punitive, the study creates no assessment stakes for students.

7. What the study can and cannot show

Can. Whether, within the same student, the occasions for productive struggle differ between writing with and without AI; what students delegate and retain; and how those patterns vary by discipline. It yields the first process-based, within-person evidence on this question from a Bangladeshi sample, and a method other institutions can reuse.

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